

Distributed Systems

Blockchain



CUHK

Satoshi Nakamoto

- Open-source 2009

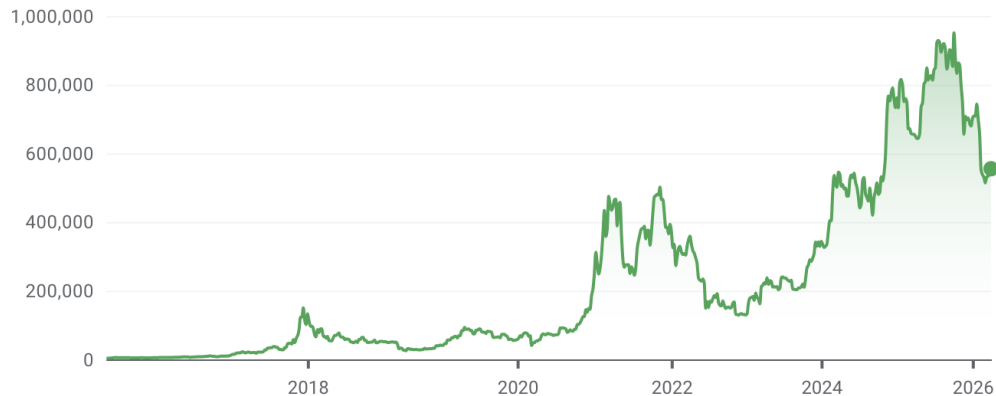
HOME > BTC / HKD • CRYPTOCURRENCY

Bitcoin to Hong Kong Dollar

555,174.41 ↑15,658.33% +551,651.36 MAX

Mar 15, 1:35:00 AM UTC · Disclaimer

1D 5D 1M 6M YTD 1Y 5Y MAX

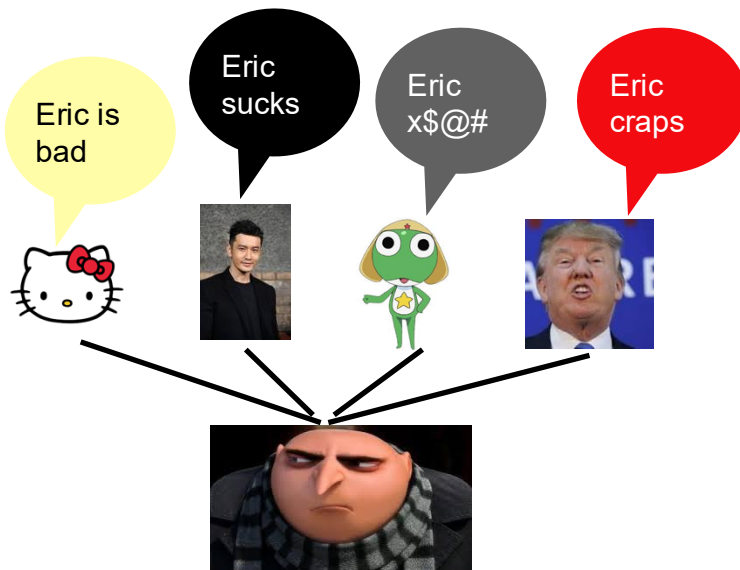


[Compare to](#)



Public blockchain is a P2P Byzantine RSM

- P2P = Decentralization
- Byzantine: *Malicious* and *Sybil*
- You can use that RSM to store whatever
 - Cryptocurrency? Store \$ transactions
 - Logistics? Store parts/goods flow

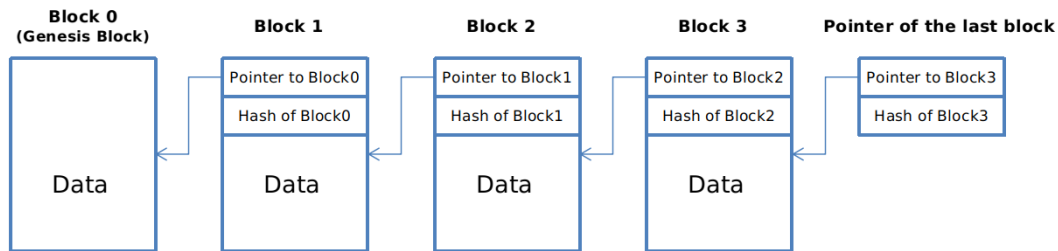
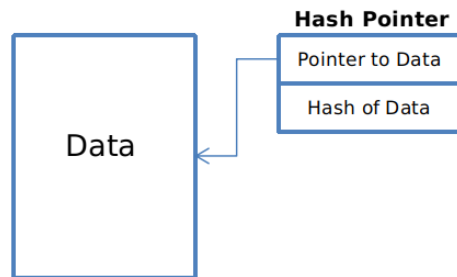
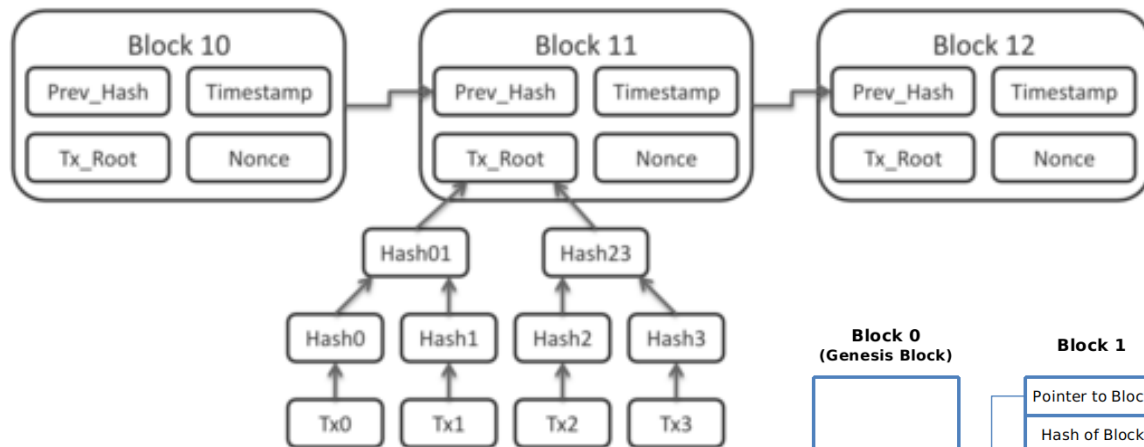


Public blockchain is a P2P Byzantine RSM

- Public blockchain is a P2P Probabilistic Byzantine RSM
 - Because you can't fully-sync under Async with 1 failure (FLP)
 - Not to mention + Byzantine
 - Not to mention + P2P scale
 - So, the majority view is the “correct” state
- Public blockchain is a P2P Probabilistic Byzantine RSM that involves two steps:
 - Leader election and Log replication!

The log is a **chain** of blocks

- Append-only
- Block-by-block
- Tamper-proof
 - A new block includes the crypto hash of the previous block



Why can we “trust” under P2P-scale Byzantine?!

- Only the leader can create a block
 - The leader can append the next block and replicate to the followers
- How to ensure the leader is honest?
 - not appending a block of transactions that transfer your \$ to my account? by cryptography
 - not appending a block that is valid but double-spending?
 - I sign \$10 to A, publish that block X to neighbors in HK
 - I sign the same \$10 to B , publish that block Y to neighbors in UK

Ensuring the leader is honest?

- By **economic + psychology**
- Randomly select the leader (luck)
- You **pay** to join the election (lottery)
- Incentive:
 - If you win the election → get mining reward (3.125 BTC since 2024; ½ since 2028)
- Why can this make the leader honest?
 - Why can this solve the sybil attack?
 - Why can this solve double-spending?
 - Who are the last ones who want Bitcoin to fail?
 - What are the odds that you can win two lotteries (blocks) consecutively?

On a new transaction is created

- You publish that to every candidate (miner)
- They will include your transaction and other transactions to form a block
- Miner:

1. Verify T is valid or not (e.g., correct signature of the owner)
2. Play a crypto puzzle (Proof-of-Work) //leader election

Brute-force finding a nonce n so that

$H(\text{PrevBlock Hash} \mid \text{Block of Txns} \mid n) < 00000000000000000000000000000888$

where H is a cryptographic hash function

Proof of Work

- $H(\dots | n) < \text{difficulty}$
 - GPU
 - FPGA
 - ASIC
 - Mining Pool



Miner: On finding a nonce ➡ win the election

- Publish the new block & nonce to everyone
- Other miners:
 - Verify that block & nonce
 - If ok, append it to my local blockchain
 - Start a new block using new transactions and start mining
 - What if a miner refuses to face the fact?

A technical summary

- Bitcoin = **Byzantine** Leader Election + **No-failure-handling** Log Replication
- The leader election is very scalable and independent of the network size
- The log replication is **way** simpler than Raft's because of inexact consensus
 - We can't control them anyway
 - They can append whatever block to the tail of their local chain...
 - So, the RSM part is just the best effort
 - No tough cases like what we see in the log replication part of Raft
 - Efficiency is also one reason
 - We trust the majority of the replicas

Beyond cryptocurrency in 2015

Using the RSM to store smartcontract1.exe



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Ether to Hong Kong Dollar

16,323.27 ↑ 1,082.51% +14,942.88 MAX

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Proof of Stake

- “skin in the game”: Validator stakes a minimum amount of Ether
- Leader election = random draw from stake distribution
- Go rogue? ”Slash the stake”

Decentralized App (Dapp), DeFi, NFT, and ICO

What is the most expensive CryptoKitties sold?

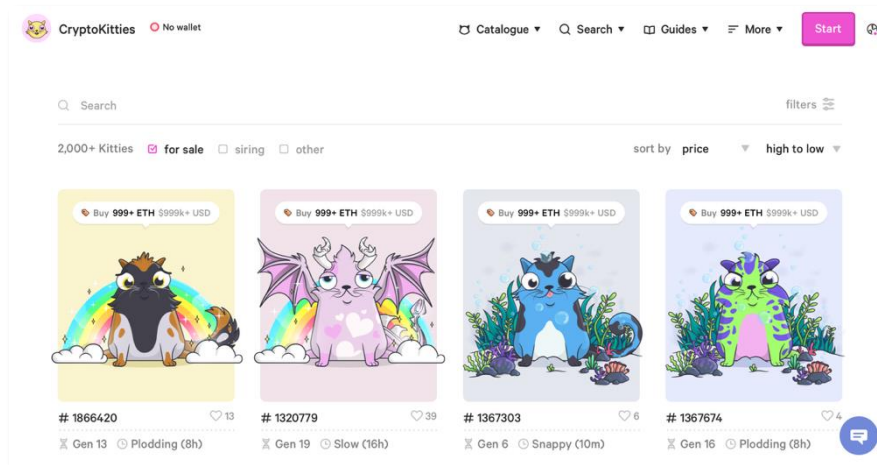
The most expensive cryptokitty ever sold was a rare "Dragon" Kitty named Dragon #18, which sold for **600 ETH (approximately \$170,000 USD)** on February 17, 2018. The buyer was an anonymous bidder. 30 Dec 2022



Quora

<https://www.quora.com/How-much-did-the-most-expens...>

How much did the most expensive cryptokitty sell for, and who bought it?



Stablecoin (ERC-20 token)

Market Summary > Tether

7.78 HKD

-0.04 (0.55%) ↓ all time

16 Oct, 12:47 am UTC · [Disclaimer](#)

1D | 5D | 1M | 6M | YTD | 1Y | 5Y | Max



Private/Permissioned/Consortium Blockchain

- No trust between business partners
- Blockchain + SmartContract as the middleman
- **Still Byzantine, but no sybil**
 - Version 1: PBFT //preventive approach
 - But not scalable to 100+ parties
- Latest Version:
 - Crash-Fault Tolerant option
 - Raft + log with signatures
 - Since **the identities are known**, the blockchain becomes an evidence in legal actions
 - Byzantine-Fault Tolerant option
 - Variant of PBFT

